

gdTAI V4.5 versus V2/V3

Frozen post-hoc comparison on the consumed 335,479-cell common benchmark | 19 August 2026

Executive conclusion

Do not promote V4.5. Its pooled highest-F1 result is inflated by 41,079 training-exposed sorted positives. On V4.5-unexposed consumed cells, highest-F1 has excessive alpha-beta FPR, while high-purity does not clearly outperform V3 balanced or V2 high-purity.

V4.5 highest-F1 pooled F1 is 0.875, but its unexposed alpha-beta FPR is **0.923%** versus **0.146%** for V3 and **0.099%** for V2 high-purity. V4.5 high-purity reduces alpha-beta FPR to **0.189%**, but unexposed F1 is 0.663 versus 0.665 for V3 and 0.753 for V2 high-purity.

Validity boundary

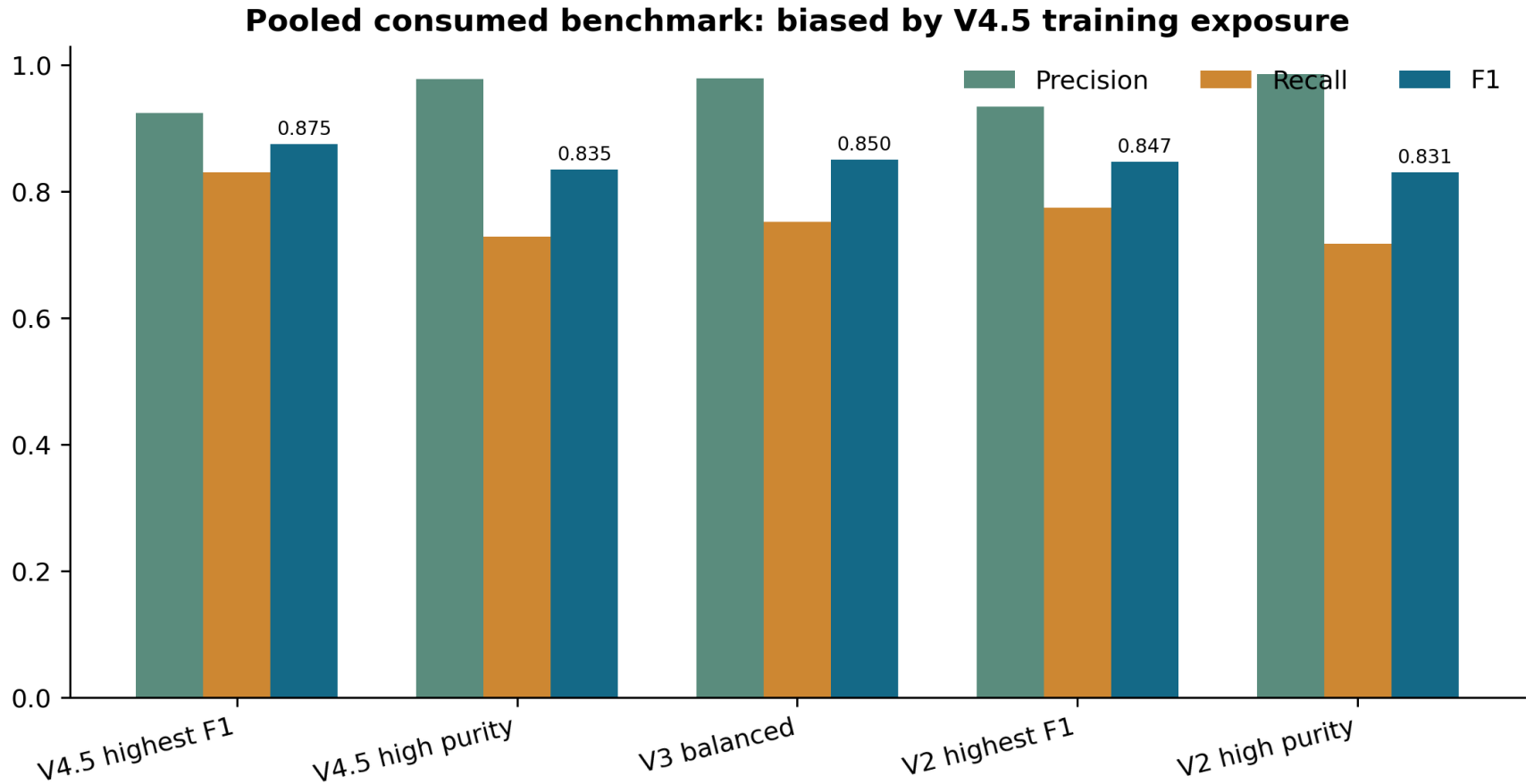
Not a new external test. Every benchmark cohort was previously inspected. GDT2020 and GDTlung were additionally used by V4.5 development.

- Training-exposed for V4.5: GDT2020 and GDTlung.
- Unexposed to V4.5 but consumed previously: BALF and eight extension sources.
- No threshold, feature, exclusion, or calibrator changed.
- No result in this report is eligible for promotion.

V4.5 exposure	Source	Cells
training_exposed_consumed	GDT_2020AUG_woCOV	25,904
training_exposed_consumed	GDTlung2023july_7p	15,175
unexposed_but_previously_consumed	BALF_BLOOD_COPD	28,328
unexposed_but_previously_consumed	GSE114724	22,283
unexposed_but_previously_consumed	GSE121636	5,954
unexposed_but_previously_consumed	GSE159251	32,420
unexposed_but_previously_consumed	GSE169246	62,695
unexposed_but_previously_consumed	GSE292700	14,801
unexposed_but_previously_consumed	GSE294273	24,137
unexposed_but_previously_consumed	GSE296954	44,371
unexposed_but_previously_consumed	GSE315928	59,411

Pooled benchmark

Pooled metrics mix training-exposed and unexposed cells. They describe prediction behavior but cannot rank generalization.

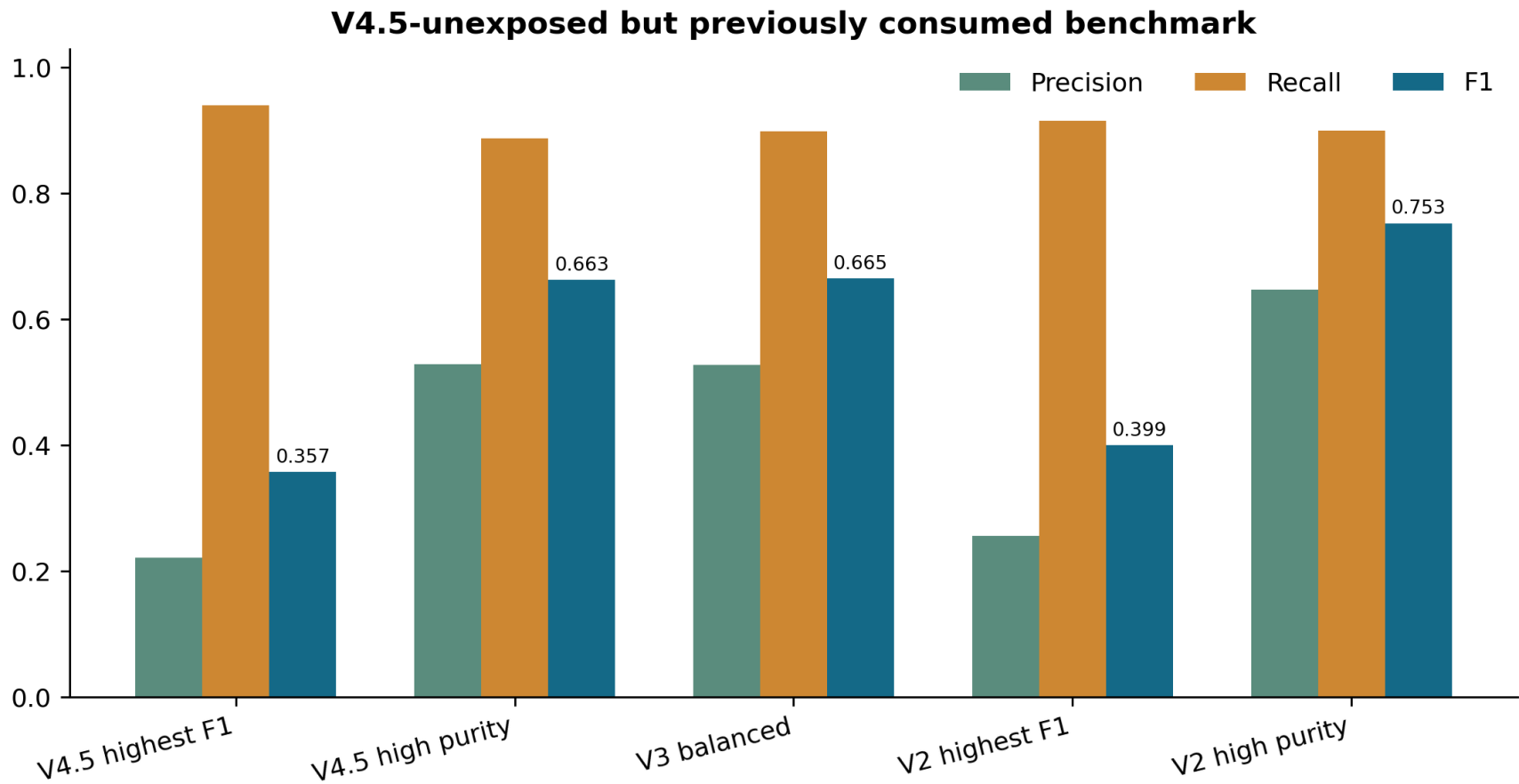


V4.5 highest-F1 benefits from its training exposure to 41,079 of 41,931 benchmark positives.

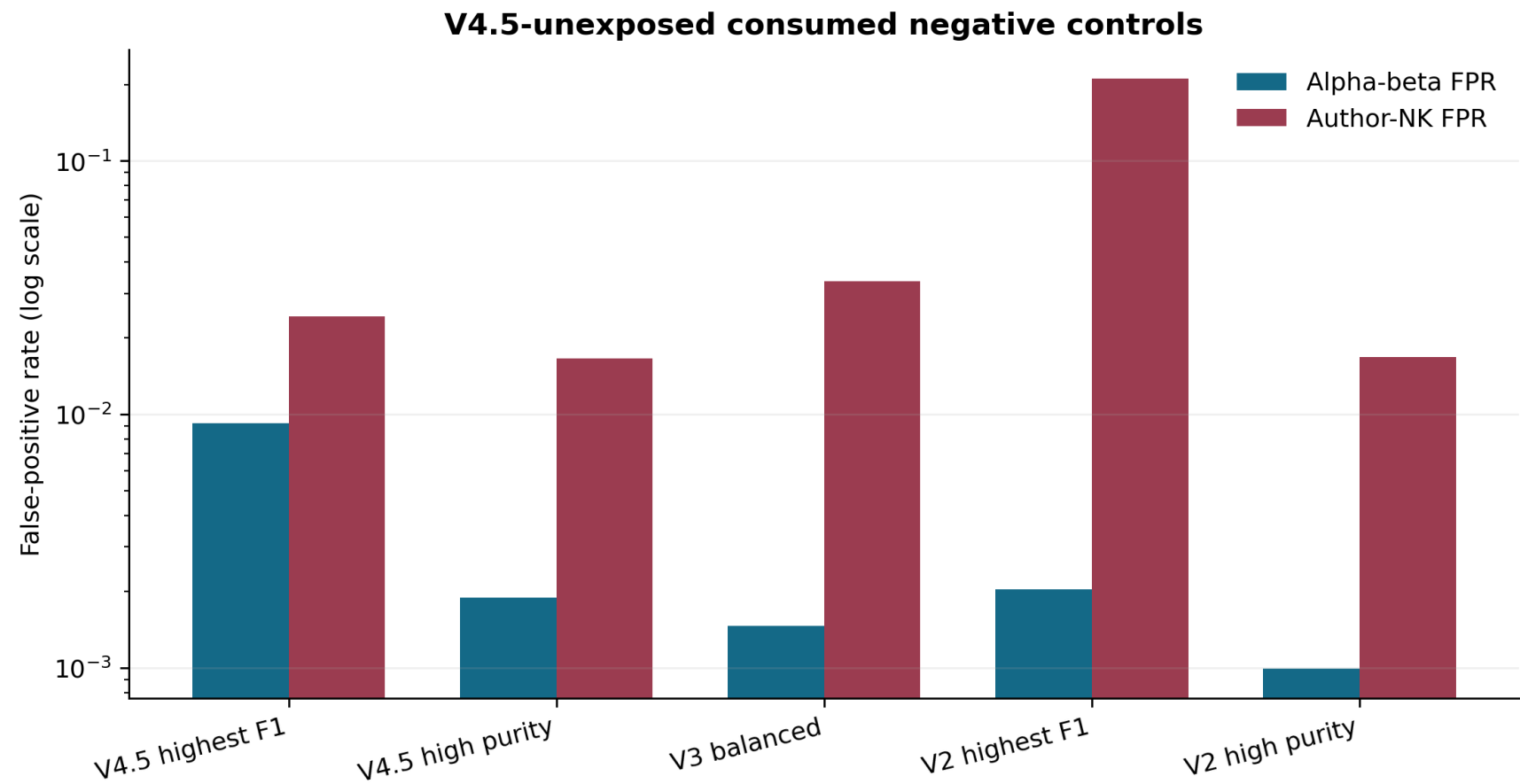
Model	Predicted	precision	recall	f1	alpha-beta FPR	NK FPR
V4.5 highest F1	37,651	92.478%	83.039%	0.8750	0.923%	2.443%
V4.5 high purity	31,212	97.844%	72.832%	0.8350	0.189%	1.658%
V3 balanced	32,210	97.870%	75.181%	0.8504	0.146%	3.352%
V2 highest F1	34,744	93.455%	77.437%	0.8470	0.204%	21.074%

Model	Predicted	precision	recall	f1	alpha-beta FPR	NK FPR
V2 high purity	30,501	98.626%	71.742%	0.8306	0.099%	1.682%

V4.5-unexposed consumed cells



This subset contains BALF as the only positive source plus all alpha-beta/NK negative extension controls.



Highest-F1 is not sufficiently specific for the very low gdT prevalence. High-purity approaches V3/V2 specificity but does not dominate them.

Model	Predicted	precision	recall	f1	alpha-beta FPR	NK FPR
V4.5 highest F1	3,633	22.048%	94.014%	0.3572	0.923%	2.443%
V4.5 high purity	1,429	52.904%	88.732%	0.6629	0.189%	1.658%
V3 balanced	1,452	52.755%	89.906%	0.6649	0.146%	3.352%
V2 highest F1	3,054	25.540%	91.549%	0.3994	0.204%	21.074%
V2 high purity	1,186	64.671%	90.023%	0.7527	0.099%	1.682%

BALF positive benchmark

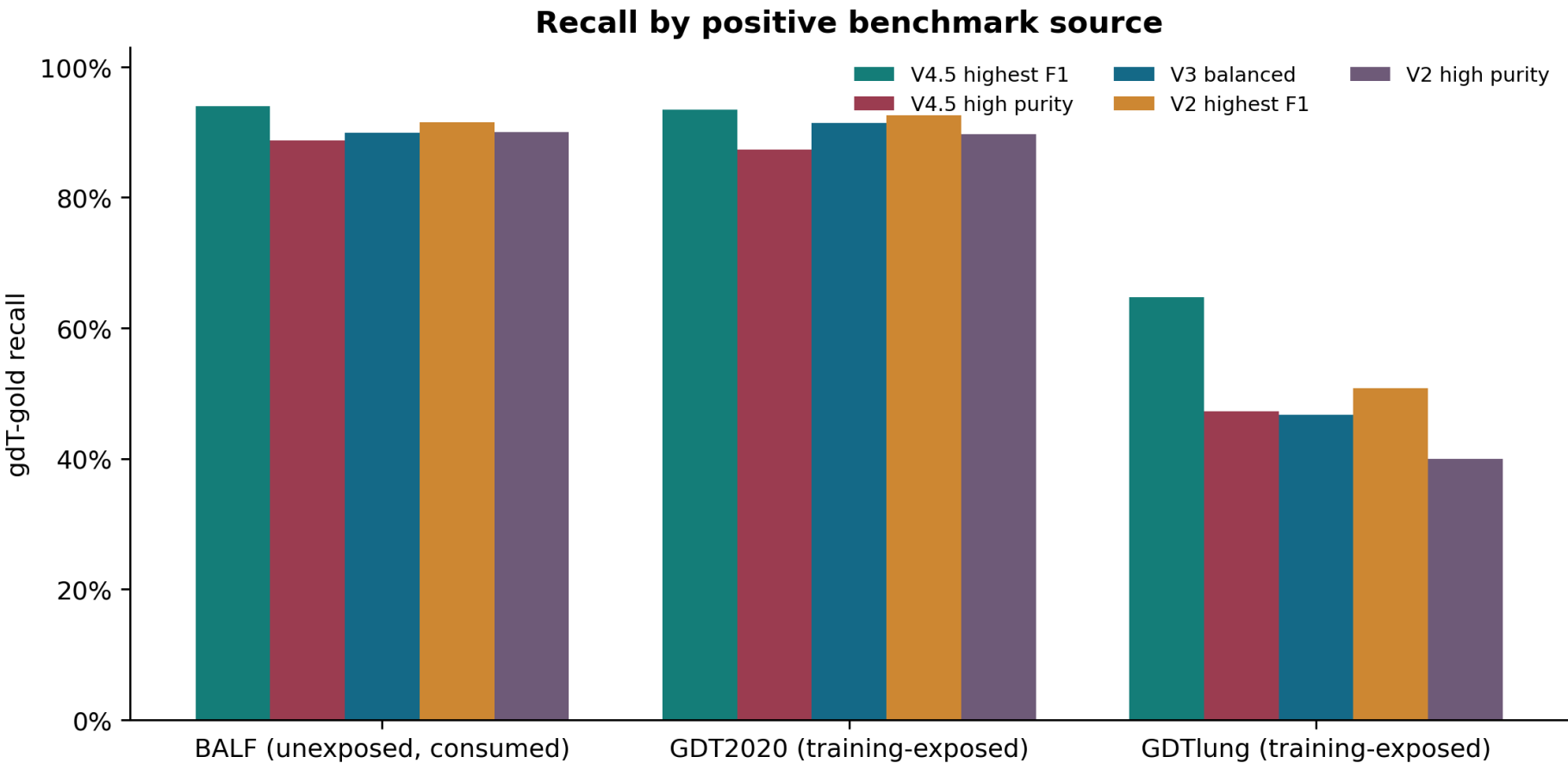
BALF was not fitted by V4.5, but it was already inspected during earlier model rounds. It is diagnostic, not external.

Model	Predicted	precision	recall	f1	alpha-beta FPR	NK FPR
V4.5 highest F1	836	95.813%	94.014%	0.9491	0.129%	0.000%
V4.5 high purity	762	99.213%	88.732%	0.9368	0.022%	0.000%
V3 balanced	776	98.711%	89.906%	0.9410	0.033%	0.394%
V2 highest F1	806	96.774%	91.549%	0.9409	0.081%	1.575%
V2 high purity	776	98.840%	90.023%	0.9423	0.033%	0.000%

V4.5 highest-F1 has the best BALF F1 (0.949) and recall 94.01%, with 0.129% alpha-beta FPR and zero observed author-NK false positives. V4.5 high-purity reaches 99.21% precision but recalls 88.73%.



Positive sources and exposure effect



Only BALF is V4.5-unexposed. Higher V4.5 recall in GDT2020/GDTlung is expected after those cohorts entered development.

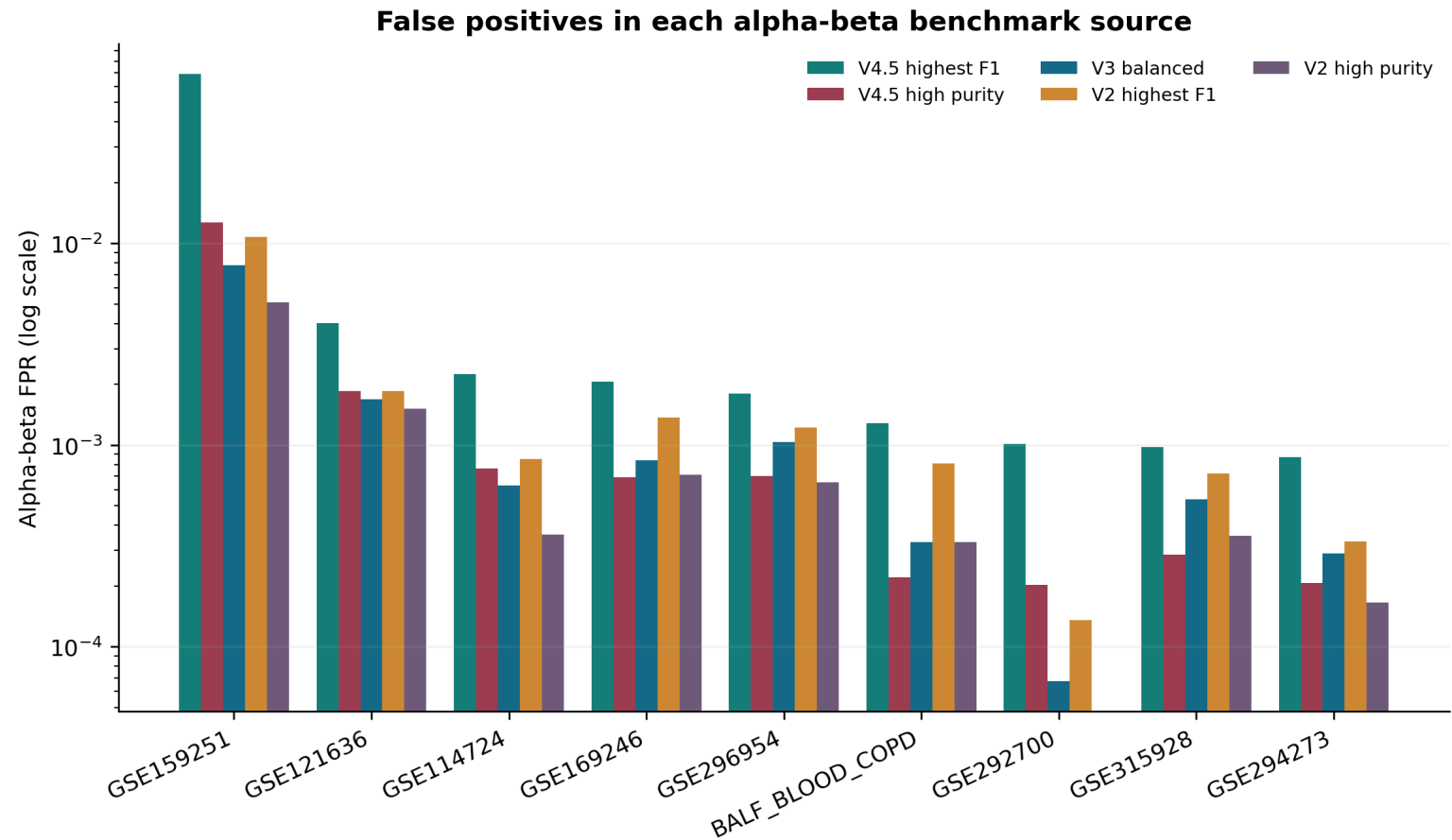
Source	V4.5 exposure	Model	gdT gold	Recall
BALF_BLOOD_COPD	unexposed_but_previously_consumed	V4.5 highest F1	852	94.014%
GDT_2020AUG_woCOV	training_exposed_consumed	V4.5 highest F1	25,904	93.422%
GDTlung2023july_7p	training_exposed_consumed	V4.5 highest F1	15,175	64.699%
BALF_BLOOD_COPD	unexposed_but_previously_consumed	V4.5 high purity	852	88.732%
GDT_2020AUG_woCOV	training_exposed_consumed	V4.5 high purity	25,904	87.330%
GDTlung2023july_7p	training_exposed_consumed	V4.5 high purity	15,175	47.189%
BALF_BLOOD_COPD	unexposed_but_previously_consumed	V3 balanced	852	89.906%
GDT_2020AUG_woCOV	training_exposed_consumed	V3 balanced	25,904	91.403%

Source	V4.5 exposure	Model	gdT gold	Recall
GDTlung2023july_7p	training_exposed_consumed	V3 balanced	15,175	46.662%
BALF_BLOOD_COPD	unexposed_but_previously_consumed	V2 highest F1	852	91.549%
GDT_2020AUG_woCOV	training_exposed_consumed	V2 highest F1	25,904	92.573%
GDTlung2023july_7p	training_exposed_consumed	V2 highest F1	15,175	50.807%
BALF_BLOOD_COPD	unexposed_but_previously_consumed	V2 high purity	852	90.023%
GDT_2020AUG_woCOV	training_exposed_consumed	V2 high purity	25,904	89.735%
GDTlung2023july_7p	training_exposed_consumed	V2 high purity	15,175	40.000%

Training-exposed pooled recall

Model	Predicted	Recall	f1
V4.5 highest F1	34,018	82.811%	0.9060
V4.5 high purity	29,783	72.502%	0.8406
V3 balanced	30,758	74.875%	0.8563
V2 highest F1	31,690	77.144%	0.8710
V2 high purity	29,315	71.362%	0.8329

Per-dataset false positives

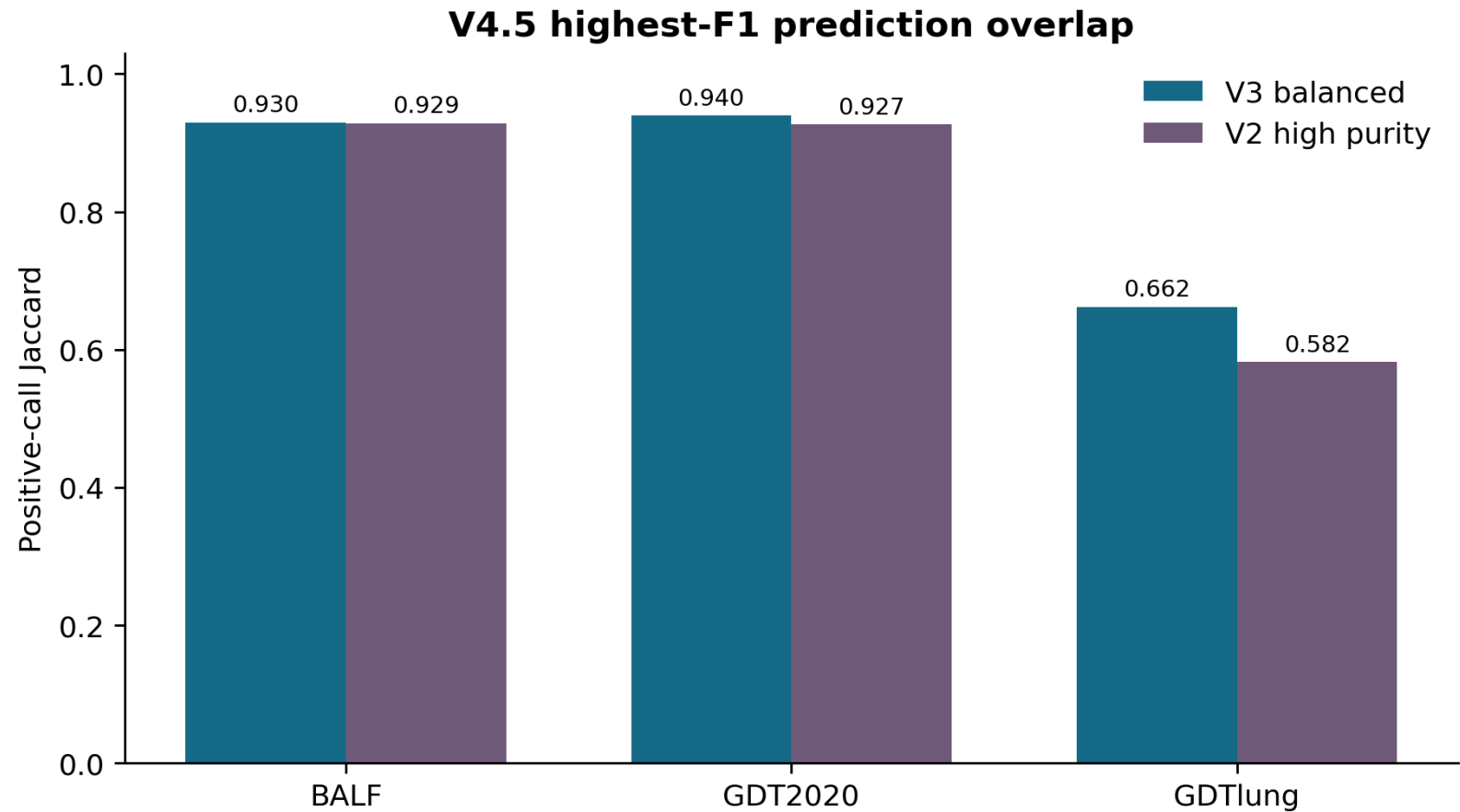


GSE159251 dominates V4.5 false positives: 6.91% for highest-F1 and 1.27% for high-purity, versus 0.78% for V3 and 0.51% for V2 high-purity.

Source	Model	alpha-beta gold	alpha-beta FPR	Author NK	NK FPR
BALF_BLOOD_COPD	V4.5 highest F1	27,222	0.129%	254	0.000%
GSE114724	V4.5 highest F1	22,283	0.224%	0	-
GSE121636	V4.5 highest F1	5,954	0.403%	0	-
GSE159251	V4.5 highest F1	32,420	6.909%	0	-
GSE169246	V4.5 highest F1	54,925	0.206%	7,770	2.523%
GSE292700	V4.5 highest F1	14,801	0.101%	0	-
GSE294273	V4.5 highest F1	24,137	0.087%	0	-
GSE296954	V4.5 highest F1	44,371	0.180%	0	-

Source	Model	alpha-beta gold	alpha-beta FPR	Author NK	NK FPR
GSE315928	V4.5 highest F1	59,411	0.098%	0	-
BALF_BLOOD_COPD	V4.5 high purity	27,222	0.022%	254	0.000%
GSE114724	V4.5 high purity	22,283	0.076%	0	-
GSE121636	V4.5 high purity	5,954	0.185%	0	-
GSE159251	V4.5 high purity	32,420	1.271%	0	-
GSE169246	V4.5 high purity	54,925	0.069%	7,770	1.712%
GSE292700	V4.5 high purity	14,801	0.020%	0	-
GSE294273	V4.5 high purity	24,137	0.021%	0	-
GSE296954	V4.5 high purity	44,371	0.070%	0	-
GSE315928	V4.5 high purity	59,411	0.029%	0	-
BALF_BLOOD_COPD	V3 balanced	27,222	0.033%	254	0.394%
GSE114724	V3 balanced	22,283	0.063%	0	-
GSE121636	V3 balanced	5,954	0.168%	0	-
GSE159251	V3 balanced	32,420	0.777%	0	-
GSE169246	V3 balanced	54,925	0.084%	7,770	3.449%
GSE292700	V3 balanced	14,801	0.007%	0	-
GSE294273	V3 balanced	24,137	0.029%	0	-
GSE296954	V3 balanced	44,371	0.104%	0	-
GSE315928	V3 balanced	59,411	0.054%	0	-
BALF_BLOOD_COPD	V2 high purity	27,222	0.033%	254	0.000%
GSE114724	V2 high purity	22,283	0.036%	0	-
GSE121636	V2 high purity	5,954	0.151%	0	-
GSE159251	V2 high purity	32,420	0.509%	0	-
GSE169246	V2 high purity	54,925	0.071%	7,770	1.737%
GSE292700	V2 high purity	14,801	0.000%	0	-
GSE294273	V2 high purity	24,137	0.017%	0	-
GSE296954	V2 high purity	44,371	0.065%	0	-
GSE315928	V2 high purity	59,411	0.035%	0	-

Prediction overlap



V4.5 agrees strongly with V2/V3 on BALF and GDT2020. GDTlung overlap is lower, showing that V4.5 gains there come from a distinct training-exposed subset.

For BALF gdT gold, V4.5 highest-F1 finds 46 cells missed by V3 but misses 11 found by V3. In GDTlung it finds 3,088 cells beyond V3 while missing 351 V3 calls; this difference cannot be interpreted as external improvement because GDTlung trained V4.5.

Model decision

- **V3 balanced remains the default.** It retains the strongest defensible balance across the consumed benchmark without V4.5's new training exposure.
 - **V2 high-purity remains the conservative fallback.** It has the lowest unexposed alpha-beta FPR and the best unexposed consumed F1.
 - **V4.5 highest-F1 is not deployment-ready.** Its GSE159251 false-positive rate is unacceptable for a rare-cell classifier.
 - **V4.5 high-purity is promising on BALF** but lacks a reproducible advantage over V3/V2 across negative sources.
 - No V4.5 threshold should be changed using this report. A new untouched positive and negative benchmark is required.
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Integrity and reproducibility

Check	Pass	Evidence
v4_5_thresholds_unchanged	1	{"high_purity": 0.9657050967216492, "highest_f1": 0.8595057725906372}
v2_v3_predictions_checksum	1	4e9157ecd6ae4368097ceb5d90cb1b6676d1ae6041c1a9d2cd89fc74e65d927b
common_feature_cache_checksum	1	b793cdbed2dbc294fabe17ef59c4c038b4f915ecbbf0d8238080b6e80c98b4ad
training_exposed_sources_labeled	1	GDT_2020AUG_woCOV,GDTIung2023july_7p
eligible_for_promotion	1	false; consumed diagnostic only

Frozen diagnostic passed. No fitting, calibration, feature selection, or threshold selection occurred; no H5AD was modified and no model was promoted or pushed.

Evaluator: workflows/gdtai/evaluate_gdtai_v4_5_vs_v2_v3_consumed.py
V4.5 contract SHA-256: 974d0687de4a93688b4e0d665b276458c90f6f5a772615dbeea9cbb2f66072b2
Input V2/V3 predictions SHA-256: 4e9157ecd6ae4368097ceb5d90cb1b6676d1ae6041c1a9d2cd89fc74e65d927b
Output predictions SHA-256: dd5696a59e19048562d0f24eb8b55d681937166d328dd7dc4f3866a44fda0ba1